BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 032 · C# Login Project · Start from Scratch

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A rising-edge-triggered D-type flip-flop has Q = 0 and D = 1. What happens to Q when the next rising edge occurs?

Q2. Why do we group neighbouring 1s together on a Karnaugh map?

Q3. Push A, then B, then C onto an empty stack. Pop once and then push D. Which item is at the top, and what is the remaining stack from top to bottom?

Q4. Using 8-bit two’s complement, write −13 and explain why simply putting 1 before the 7-bit magnitude is not correct.

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I Do · Read new knowledge

Key vocabulary

sequence — Statements executed in order.

selection — Branching with if/else.

iteration — Repeating statements with a loop.

validation — Checking input meets defined rules.

array — Indexed fixed-size collection, possibly multidimensional.

record — Named fields grouped as a single item.

list — Collection that can grow or shrink.

New learning

What must a login program do? — Read two inputs, decide whether both match the demonstration account, and show a clear result. Design the sequence before typing code.

Why start with sequence? — Read username; read password; display what was entered only during debugging. In the final version never echo a password.

Introduce selection — Use AND (&&) to require BOTH username and password to match. One wrong value must mean access denied.

Make a second attempt possible — A loop can repeat after invalid credentials. Add a maximum number of attempts to avoid an endless loop.

Data-structure revision — Write one accurate four-option MCQ each on arrays (up to 3D), records, lists and tuples. Add answers and distractor explanations.

Worked example

Model the first three lines — not the finished program

1. Create a console program — CODE:

```
using System;

Console.Write("Username: ");
string username = Console.ReadLine() ?? "";
```

2. Add the second input — CODE:

```
Console.Write("Demo password: ");
string password = Console.ReadLine() ?? "";
```

3. Plan, then add selection — CODE:
// Write an if statement:
// username must equal "student"
// AND password must equal "demo123"
// Print "Access granted" or "Access denied".

We Do · Practise with guidance

Build the condition

What Boolean operator makes BOTH checks compulsory?

Explain your solution first

```
if (username == "student" && password == "demo123")
{
    Console.WriteLine("Access granted");
}
else
{
    Console.WriteLine("Access denied");
}
```

Handle repeated attempts

How could you allow at most three attempts?

```
for (int attempt = 1; attempt <= 3; attempt++)
{
    // Read inputs, check both values, break on success.
}
```

MCQ quality check

A student writes: 'Which structure is best?' Why is this too vague?

Specify the operation, such as key-based lookup, FIFO removal or index access, so there is one defensible correct answer.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

CODING 1 · SEQUENCE
From a blank C# console program, request a username and a demonstration password. Use two string variables. Do not copy in a finished solution.

Question 2

CODING 2 · SELECTION
Add an if/else. Grant access only for username 'student' AND demo password 'demo123'. Test correct name/wrong password and vice versa.

Question 3

CODING 3 · ITERATION
Extend the program to allow at most three attempts. Show 'Locked for this session' after three unsuccessful attempts.

Question 4

SYNOPTIC MCQ BANK
Write FOUR original four-option MCQs: one on arrays up to 3D, one on records, one on lists and one on tuples. Mark the correct letter, explain it and justify each distractor.

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Exit Ticket · Answer before checking

A login should allow access when the username is correct even if the password is wrong.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Q becomes 1 at the rising edge because the flip-flop samples D. It then holds that value until the next triggering edge.

Q2. We group neighbouring 1s to simplify the Boolean expression. Variables that change within the group can be removed.

Q3. D is at the top. From top to bottom the stack is D, B, A. C was removed by the pop.

Q4. 13 = 00001101_2. Invert to 11110010_2, then add 1: 11110011_2. Prefixing the magnitude gives sign-and-magnitude (10001101_2), not two’s complement.

You Do pages · Independent questions

Q1. CODING 1 · SEQUENCE
Model: using System; read the username and demonstration password using Console.ReadLine() ?? "". The task is complete only when the program compiles and handles both inputs. Never use real passwords.

Q2. CODING 2 · SELECTION
if (username == "student" && password == "demo123") Console.WriteLine("Access granted"); else Console.WriteLine("Access denied");. AND is essential: one matching value alone is insufficient.

Q3. CODING 3 · ITERATION
for (int i = 0; i < 3; i++) { read both credentials; if (both match) { grant access; break; } if (i == 2) display lock message; }. Test successful attempt 1, success attempt 3 and three failures.

Q4. SYNOPTIC MCQ BANK
MCQ exemplars: 3D array arr[2,3,4] holds 24 entries; record groups named fields; a list allows a variable-length collection; a tuple groups a fixed set of possibly different types. Each authored MCQ must identify one correct response and justify distractors.

Exit Ticket · Model answer

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False. A login should check both the username and password before allowing access.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: